



Department of Science and Technology
Forest Products Research and Development Institute



Bamboo Musical Instruments Processing Center





BMIPC



The Bamboo Musical Instruments Processing Center (BMIPC) houses a line of processing equipment for BMI manufacturing purposes or for businesses that wish to adopt the Institute's technologies for improving the processing, durability, aesthetics and sound quality of select BMIs. The BMIPC also features a mini museum showcasing instruments developed in the Center and other BMIs made by several commercial makers and indigenous groups in the country. It will serve as a venue for training teachers, students, hobbyists or interested adopters on making and tuning of BMIs.

SERVICES OFFERED

01

Technical Assistance



02

Advisory Services



03

BMI Tuning and Making



04

BMI Processing



05

Mini Museum Tours



BMI Prototypes

Thru the BMI PC, the DOST-FPRDI has fabricated several bamboo musical instrument prototypes that include angklung, marimba and kawagong.



These instruments were crafted using select processing equipment at the Center. To ensure the BMIs' durability without compromising its tonal quality, the DOST-FPRDI carefully studied and applied drying, finishing & preservation techniques.

Processing



Treatment Vats

For soaking bamboo in treatment chemicals.



Bamboo Splitter

For faster splitting of bamboo into several pieces.



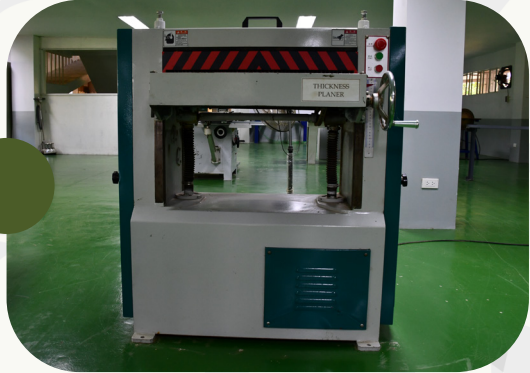
Bamboo Pole Cross-Cutter

For cutting bamboo poles to desired length.

Equipment

Thickness Planer

For trimming bamboo to a consistent thickness throughout their length and flat on both surfaces.



Radial Arm Saw

For ripping, cutting bevels or miters, dados and rabbets, form moldings, and serving as guide for a router.

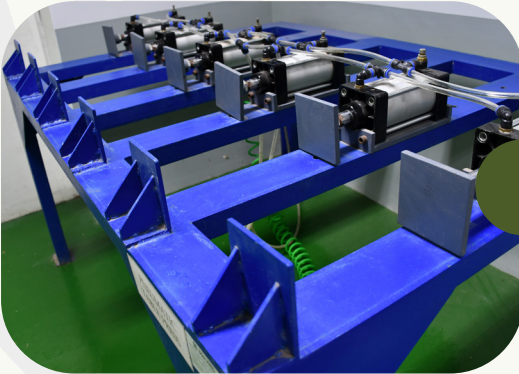


Table Saw

For making straight or angled cuts through moderately soft materials such as bamboo, wood or similar materials.



Processing



Pneumatic Table Press Clamp Machine

For holding or securing glued bamboo materials tightly together to prevent movement or separation through the application of inward pressure.



Four-Sided Planer

For executing the following woodworking: board, straightening on four sides, planing on four sides eliminating the crooked/ raw part of the wood/ bamboo perfect board, removing wood imperfections.



Unit Jointer/ Surface Planer

For straightening the edges of boards in the operation known as jointing . A jointer plane may also be used to flatten the place of a board.

Equipment

Finishing Spray Booth

For containing paint overpray, keeping your employees and environment safe.



Bench-Type Drill Press

For drilling, but other possible operations include: raming, countersinking, counter-boring, and tapping.

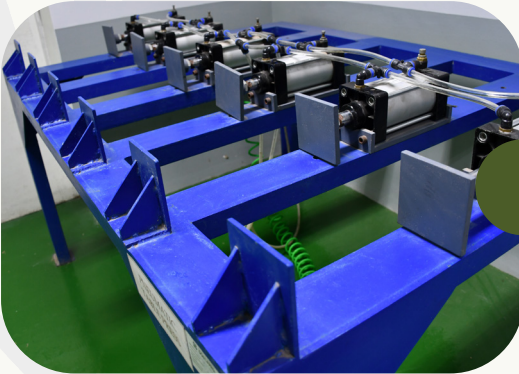


Electric Drying Chamber

For controlling humidity in storing and dehumidifying test objects.



Processing



CNC Lathe

For turning, facing, grooving, threading, grooving and parting operations can be done on cylindrical workpiece to make desired profile product.



Twin Rip Saw

For making a rip cut, a cut made parallel to the direction of the wood grain.



Finishing Tools

For finishing products produced.

Equipment

Dowel Machine

For cutting bamboo into round pieces. It is a specialized woodworking tool which uses a rotating drill bit to accurately drill holes in wood.



Standing Drill Press

For drilling precise holes in wood or metal.



Sanding Machine

For sanding wood. It is a stationary machine that consists of a replaceable circular shaped sandpaper attached to a wheel.



Processing



Oscillating Belt Sander

For sanding wood. It is mounted on a spindle that both rotates and oscillates in and out or up and down along the axis of the spindle.



Belt and Disc Sander

For trimming to a scribed line and sanding very rough leveling surfaces.



Industrial Dust Collector

For collecting dust. It consists of a blower, a filter-cleaning system, dust filter and a dust receptacle.

Equipment

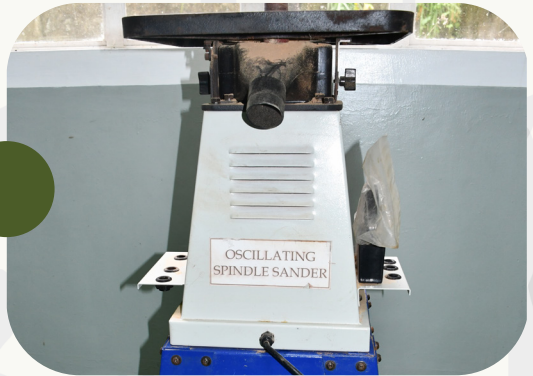
Non-Vibrating Table

For reducing vibration disturbances caused by nearby equipment. These balance accessories can also reduce shock when using precision testing equipment.



Oscillating Spindle Sander

For sanding edges of boards. This oscillating spindle sander is a benchtop (or in some commercial models, floor-standing) woodworking tool. The sander has a small tabletop and a sanding drum that protrudes up through the rear portion of the tabletop.



Buffing Machine

Used for polishing and smoothing surfaces for a more refined and less dull look.



BMI TECHNOLOGIES



Seasoning or Drying

Seasoning or drying is the process of removing moisture or water from bamboo to improve its serviceability and suitability. Air drying is extensively done around the country by local BMI makers, but it takes months before it dries to equilibrium moisture content (EMC).

FPRDI is developing a dehumidifying dry kiln for BMI makers. This technology uses heat to remove moisture and recycle the heat continuously making it more efficient than traditional dryers.

Each bamboo species has its own kiln drying schedule to follow. Kiln drying schedule is carefully worked-out compromise between the need to dry bamboo as fast as possible and at the same time, to avoid severe drying conditions that will cause drying defects. It is a series of dry- and wet-bulb temperature that establish the temperature and relative humidity in the kiln and are applied at various stage of drying process. The schedules must be developed so that drying stresses do not exceed the strength of bamboo at any given temperature and moisture content. Otherwise, the bamboo pole will crack either on the surface or internally, or be crushed by forces that collapse the bamboo cells.



Preservation

Preservation is the process that involves the application of chemical preservatives into the wood and non-wood forest products to prevent deterioration. It is necessary to manage post-harvest pests through non-chemical and chemical treatment applications to maintain the quality of raw materials and finished products.

Thermal Modification

Thermal modification is the process of heating a material to a high temperature (at least 150°C) without the presence of oxygen inside the system. It is a modification process which aims to address various drawbacks of bamboo as a biological material, making it dimensionally stable (less hygroscopic) and less susceptible to biological attacks.



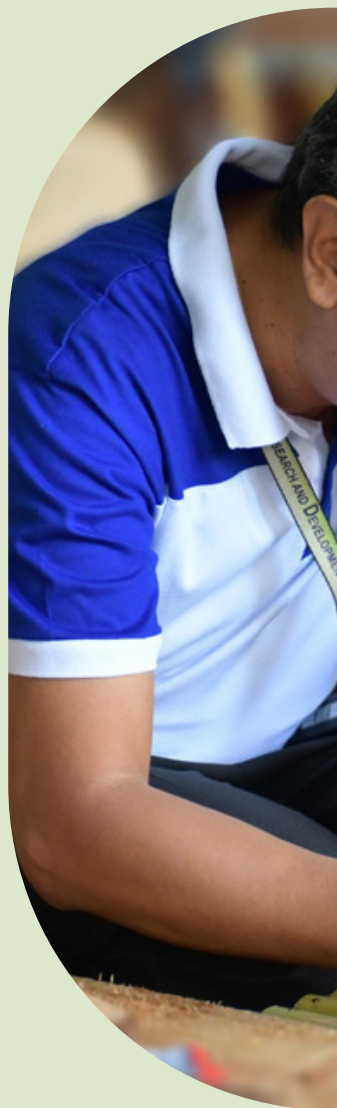
Finishing

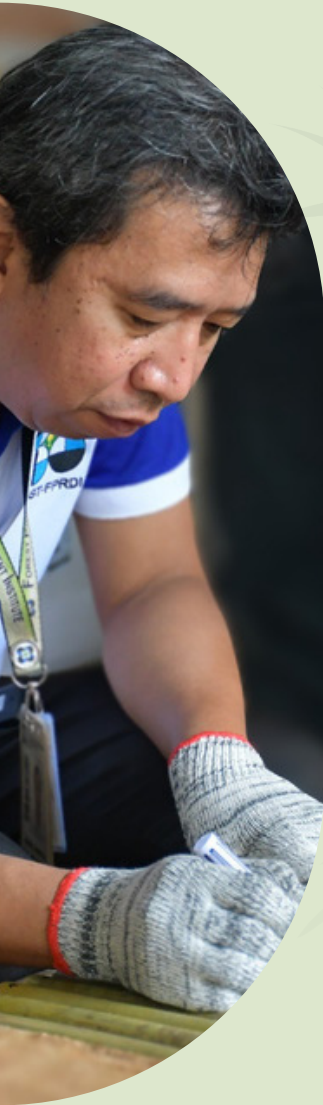
Finishing is the final step in the construction of bamboo musical instruments. This increases the overall appeal and aesthetics of BMI and provides additional protection. A musical instrument must be finished in such a way that it will not affect the tune and vibration of sound. Finishing involves the process of applying a transparent or semi-transparent liquid coating onto the substrate, in this case, bamboo, to emphasize its natural beauty with or without altering its natural color.



Apply as an incubatee under the DOST- FPRDI Technology Business Incubator and have access to:

- BMI Processing Center and Technologies
- Wood and non-wood processing technologies
- Networking and partnerships
- Technology trainings
- Business monitoring
- Marketing and finance monitoring
- Assistance in intellectual property protection
- Shared operational costs
- Office space rental
- Venture capital firms and investors





Application Process:

- 1** Requirements Submission
- 2** Business Model Presentation
- 3** Applicant Screening
- 4** MOA Signing



Requirements:



Application letter

*Send to info@fprdi.dost.gov.ph addressee:
FPRDI Director Dr. Romulo Aggangan*



Duly accomplished FPTBI application form

Request copy from info@fprdi.dost.gov.ph



Business plan/ business model canvas/ feasibility study



Mayor's business permit



Copy of any valid ID



Statement of accounts



Applicant screening

Who are eligible to apply:

Sole Proprietorships, Partnerships, Corporations,
and Cooperatives

Business Startups

Micro, Small and Medium Enterprises (MSMEs)

Graduating Students, Faculty, and Researchers

For more information, please contact:

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